DAYANANDA SAGAR UNIVERSITY

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Bachelor of Technology in
COMPUTER SCIENCE AND ENGINEERING

Major Project Phase-II Report

Maximising Section Throughput Using AI-Powered Precise Train Traffic Control

Batch: 2022-2026

By

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(2025-2026)

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CERTIFICATE

This is to certifythat the Major Project Stage-I work titled **"MAXIMISING SECTION THROUGHPUT USING AI-POWERED PRECISE TRAIN TRAFFIC CONTROL"** is carried out by **Pemmasani Hima Chandana (ENG22CS0389), Rahul Kumar Sahoo (ENG22CS0404), Rethi Reckwin J E (ENG22CS0410), Prakhar Sinha (ENG22CS0571),** bonafide students seventh semester of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025- 2026**.

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DECLARATION

We, **Pemmasani Hima Chandana (ENG22CS0389), Rahul Kumar Sahoo (ENG22CS0404), Rethi Reckwin J E (ENG22CS0410), Prakhar Sinha (ENG22CS0571),** are students of seventh semester B. Tech in **Computer Science and Engineering**, at School of Engineering, **Dayananda Sagar University**, hereby declare that the Major Project Stage-I titled **"MAXIMISING SECTION THROUGHPUT USING AI-POWERED PRECISE TRAIN TRAFFIC CONTROL"** has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science and Engineering** during the academic year **2025-2026.**

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ABSTRACT

The Indian Railways network handles one of the world's largest passenger and freight movements, yet relies extensively on manual decision-making for traffic management. This leads to delays, congestion, and reduced operational efficiency. This project proposes a Machine Learning–based predictive system that forecasts train delays, congestion levels, and Section Occupancy Time using historical and synthetic datasets. The proposed solution generates alerts, provides dashboard-based visualization, and assists Traffic Controllers by offering data-driven recommendations. The model aims to enhance punctuality, reduce congestion, and optimize routing across railway sections. The system integrates data preprocessing, feature engineering, and machine learning techniques such as Logistic Regression and Random Forest to achieve reliable predictions. By combining predictive analytics with simulation and visualization, the proposed solution supports proactive decision-making in railway traffic management. Overall, the system demonstrates the potential of AI-driven approaches to improve operational efficiency and reliability in large-scale railway networks.

Chapter 1 – INTRODUCTION

1.1 BACKGROUND

Indian Railways is one of the largest transportation networks in the world, carrying millions of passengers and freight containers every day. Its operations span thousands of kilometers of track divided into numerous block sections, each requiring precise coordination to ensure smooth and safe train movement. Despite its massive scale and technological advancements in signaling and safety systems, the decision-making within railway traffic management is still largely reliant on the experience and intuition of Train Traffic Controllers. As train frequency continues to increase and network congestion becomes more common, the manual approach faces significant limitations. Delays spread across interconnected routes, leading to cascading effects that impact punctuality, operational efficiency, fuel usage, and customer satisfaction. To address these challenges, there is a growing need for intelligent, data-driven systems capable of predicting train delays, estimating congestion, and assisting controllers in making faster and more accurate decisions.

1.2 MOTIVATION

The motivation for this project stems from the need to strengthen existing railway traffic operations through data-driven predictive analytics, especially in complex and dynamic environments. In railway networks, a delay affecting one train can influence subsequent trains entering the same block section, creating a chain reaction that impacts overall network efficiency. While current systems provides tracking and management, there is growing value in complementing them with intelligent tools that can generate predictive insights on future delays, likely congestion patterns, and section occupancy times. By integrating machine learning-based forecasting with existing operational mechanisms, the system aims to support controllers in making more informed and proactive decisions. This approach enhances situational awareness, optimizes schedule adherence, and contributes to improved operational planning and network reliability.

1.3 OBJECTIVES

The primary objective of this project is to develop a machine-learning–based predictive framework that enhances decision-making for Train Traffic Controllers by accurately forecasting upcoming delays, estimating section-wise congestion levels, and computing block occupancy times. The system utilizes supervised learning models such as Logistic Regression, Random Forest, and XGBoost, supported by advanced feature engineering derived from timetable metadata, sectional attributes, maximum permissible speeds, gradients, line capacities, priority classifications, weather parameters, and historical movement patterns. A domain-aligned dataset has been structured to enable multivariate analysis, cross-validation, and performance evaluation through metrics such as accuracy, MAE, RMSE, and R^2. The final objective is to integrate these predictive models into an interactive analytics dashboard that visualizes delay patterns, congestion heatmaps, and occupancy timelines, providing data-driven, algorithmically supported insights to improve the efficiency, safety, and operational responsiveness of railway traffic management.

1.4 SCOPE

The scope of this project includes modelling railway traffic movement for a selected zone or route, generating synthetic datasets that mimic actual behavior, and building machine learning models to predict delays and congestion. The system focuses on passenger trains, express trains, superfast trains, and freight trains, each of which contributes differently to congestion. While the project captures key operational factors such as block sections, halt durations, signal delays, speed restrictions, and weather conditions, it does not attempt to model the entire Indian Railways network due to its complexity. Instead, the project uses a representative portion of the network to demonstrate the feasibility and effectiveness of the proposed solution. The scope also includes the development of a dashboard for visualizing train movement predictions.

Chapter 2 – PROBLEM DEFINITION

The core focus of this project is to enhance railway operations by integrating predictive analytics into existing decision-making processes. Traffic Controllers currently make operational decisions such as granting signals, planning crossings, managing section occupancy, and handling train diversions by combining available system outputs with their expertise and situational awareness. However, as train density increases and operational complexity rises, even minor delays can propagate through interconnected sections and impact multiple subsequent movements. To support efficient planning, there is significant potential in augmenting existing mechanisms with intelligent, machine learning–based forecasting tools that can estimate future delays, anticipate congestion levels, and predict section traversal times. By offering early insights and actionable predictions, the system strengthens proactive decision-making, optimizes resource utilization, and contributes to smoother railwayoperations and enhanced passenger experience.

` Chapter 3 – LITERATURE SURVEY

3.1 LITERATURE SURVEY

Ref Number	Citation	Summary
1	European Transport Research Review, 2025	Provided a systematic review ofexisting ML and statistical models; highlighted gaps such as limited interpretability, lack of generalizability, and privacy concerns
2	IEEE Access, 2024	Achieved ~95% accuracy in predicting delays up to 15 minutes in advance; validated with real-world railway OCC data
3	IARJSET,2021	The RF model showed the highest prediction accuracy
4	IET Intelligent Transport Systems, 2023	XGBoost, combined with metaheuristic optimisation, improves prediction accuracy and handles dispatch commands effectively.
5	Applied Soft Computing, 2021	XGBoost with Bayesian Optimisation provides improved accuracyfor arrival delay prediction.
6	Recent Advances in Computer Science & Communications, 2019	Weather-based delay modelling shows Random Forest and ANN outperform simpler models.
7	Journalof Advanced Transportation, 2021	The hybrid ELM-PSO model effectively captures delay patterns and performs better in nonlinear scenarios.
8	Transportation Research Part C, 2021	Hybrid rule-based + ML model integrating operational constraints for more interpretable delay prediction.
9	Applied Soft Computing, 2022	Spatio-Temporal Graph Neural Network (ST-GNN) predicts delay propagation across network topology.

10	European Transport Research Review, 2023	Review of ML for railway analytics; identifies challenges in deploying models at scale in real-world rail operations.
11	IEEE ITSC Conference, 2019	Generalised delay prediction system for urban railways using historical data and ML techniques.
12	Expert Systems With Applications, 2021	A cost-sensitive CNN framework treats timetables as images for delay propagation pattern recognition.
13	IEEE Access, 2019	Data-driven train dispatching management review; emphasises ML for delay distribution and rescheduling.

Chapter 4 – PROJECT DESCRIPTION

4.1 PROPOSED DESIGN

The proposed system architecture consists of five major layers, each performing a critical function in processing railway operational data and generating actionable insights for effective decision-making.

- **Input Layer:**
  This layer utilizes train metadata, timetables, speed profiles, locomotive types, weather conditions, and historical congestion records to serve as the foundational data input for the system.

- **Preprocessing Layer:**
  In this layer, data cleaning and feature engineering are performed, followed by the construction of the route graph to model train movements across sections.

- **Prediction Layer:**
  This layer employs machine learning models, particularly the Random Forest algorithm, to predict train delays, congestion levels, and other operational parameters.

- **Simulation Layer:**
  The simulation layer focuses on calculating section occupancy and simulating train movement to analyze operational flow under different conditions.

- **Visualisation Layer:**
  This layer displays the predicted delay, section throughput, and other insights through graphical and interactive visualizations to assist decision-makers effectively.

Block Diagram:

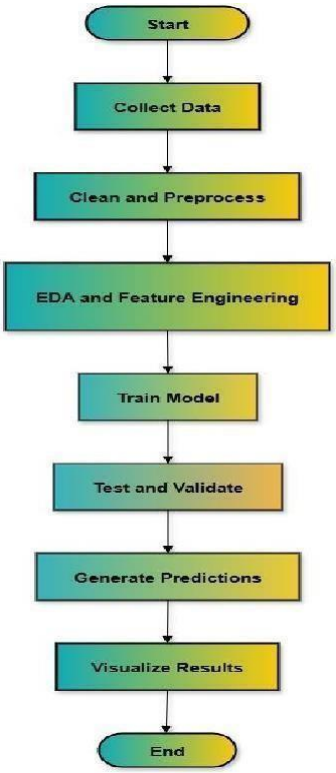


Figure 4.1(a): Proposed Block Diagram

- **Start**

  This block marks the beginning of the machine-learning workflow. It indicates that the system is ready to initiate the process of building a prediction model, starting from raw data and moving towards generating insights.

- **Collect Data**

  In this stage, all relevant datasets are gathered from different sources. This may include train schedules, delay logs, weather information, and route congestion data. The purpose is to bring together all the raw inputs required for building an accurate prediction system.

- **Clean and Preprocess**

  Raw datasets usually contain missing values, inconsistent formats, and duplicate records. This step focuses on correcting these issues to ensure high data quality. It involves handling gaps, converting data types, encoding categories, and preparing the dataset so that algorithms can process it effectively.

- **EDA and Feature Engineering**

  Exploratory Data Analysis helps uncover patterns, trends, and relationships within the dataset. It provides insights into factors influencing delays. Feature engineering then transforms those insights into new, meaningful features, such as time-based fields, weather indicators, or historical averages, improving the model's predictive strength.

- **Train Model**

  Here, machine-learning algorithms learn from the prepared training data. Models like Logistic Regression or Random Forest analyze patterns in the input features and learn how these patterns relate to train delays. The outcome is a trained model that begins to understand delay behavior.

- **Test and Validate**

  The trained model is evaluated using unseen data to measure how well it performs. Metrics like accuracy, MAE, RMSE, and R^2 score help determine whether the model generalizes properly. This step ensures that the predictions are reliable and not overfit to the training data.

- **Generate Predictions**

  Once validated, the model is used to predict delays for new train data. It takes current features such as time, weather, and congestion and produces an estimated delay or a classification indicating whether the train will be late**.**

- **Visualise Results**

  Visualisations help clarify the model's performance and predictions. Graphs such as feature importance charts, actual vs predicted delay plots, and accuracy comparisons make it easier to understand the outcomes and present them in reports or dashboards.

- **End**

  This marks the completion of the pipeline. At this stage, the system has successfully transformed raw data into insights and predictions, enabling better decision-making and analysis of train delay behavior.

4.2 ASSUMPTIONS & DEPENDENCIES

• Trains operate on fixed blocks.

The system assumes that railway tracks are divided into predefined blocks, and only one train can occupy a block at a time. This ensures safe movement and predictable scheduling, making it easier to model delays and train interactions within the network.

• Section length and permissible speeds are known.

It is assumed that the length of each track section and its maximum allowed speed are predefined and available. These values are essential for calculating travel time, estimating delays, and simulating how trains move through different parts of the route.

• Weather influences speed.

The model depends on the assumption that weather conditions affect train speeds—for example, heavy rain, fog, or storms may cause slower movement. This helps integrate weather-based features and justifies using weather data for delay prediction.

• Priority trains override normal scheduling.

Some trains, such as express, premium, or emergency services, are assumed to have higher priority. This means they may get precedence over other trains when allocating blocks or handling congestion, which affects the behaviour of delays and routing.

• Synthetic data simulates realistic train movement patterns.

Where or historical data is incomplete, synthetic datasets are assumed to represent realistic train operations. This helps in training and testing models by mimicking actual behaviour such as speed variations, delays, and congestion patterns.

Chapter 5 – REQUIREMENTS

5.1 FUNCTIONAL REQUIREMENTS

The system has six major functional requirements, each designed to ensure accurate delay prediction, efficient traffic flow management, and effective operator support within the railway network

.

- **Predict the upcoming delay for each train.**

  The system should be able to analyse the input features, such as schedule, speed, weather, and past performance, to forecast how late a train is expected to be at upcoming stations. This enables proactive planning and supports better operational decisions.

- **Estimate congestion of each section.**

  The model must assess how many trains are occupying or approaching a section at a given time. By analysing block occupancy and predicted movement, the system identifies bottlenecks and provides a clear view of the current network load.

- **Generate Section Occupancy Time.**

  For each train, the system should compute how long it will take to pass through a particular section. This helps in determining block availability, optimising routing, and avoiding conflicts where multiple trains need the same track segment.

- **Provide dashboard visualisation.**

  A user-friendly dashboard must be available to display train delays, congestion levels, section occupancy, and model performance metrics. Visual graphs and charts help operators quickly understand system status and make informed decisions.

- **Allow simulation of "What-if scenarios".**

  The system should support hypothetical scenarios such as speed reductions, weather changes, or priority adjustments. This allows users to test how different conditions would affect delays, capacity, and overall traffic flow in the network.

- **Generate alerts for high congestion.**

  The model must trigger warnings when congestion in any section crosses predefined thresholds. These alerts help operators act quickly to manage traffic, reroute trains, or take preventive measures to avoid delay escalation.

5.2 NON-FUNCTIONAL REQUIREMENTS

The system defines several key non-functional requirements to ensure strong performance, reliability, and maintainability beyond its core predictive capabilities. It targets a minimum prediction accuracy of 85%, ensuring dependable and trustworthy results that operators can rely upon in decision-making. Additionally, each prediction should be generated in less than two seconds, providing near responsiveness, which is critical in dynamic and fast-paced operational environments.

To support long-term usability, the system is built using a modular architecture that enables easier maintenance, component upgrades, and seamless integration with existing railway infrastructure. Furthermore, the solution is designed for high scalability, allowing it to efficiently accommodate growing data volumes, an increasing number of users, and expansion of the railway network without compromising overall performance.

Chapter 6 – METHODOLOGY

6.1. DATASET CREATION

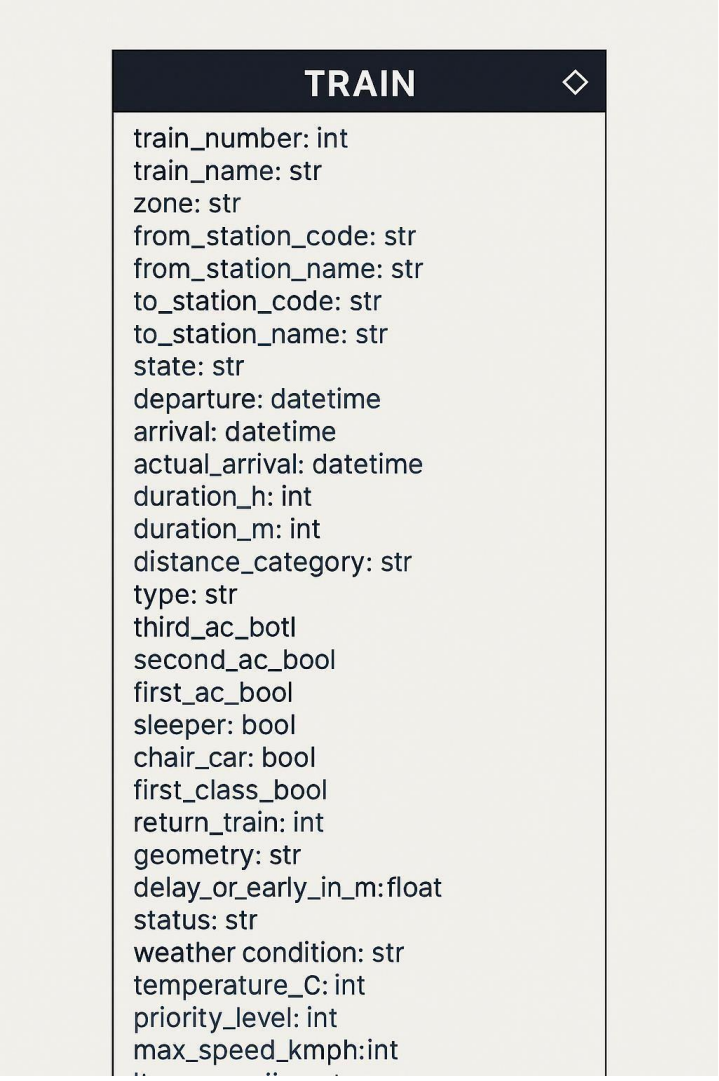


Figure 6.1(a): Dataset Class Diagram

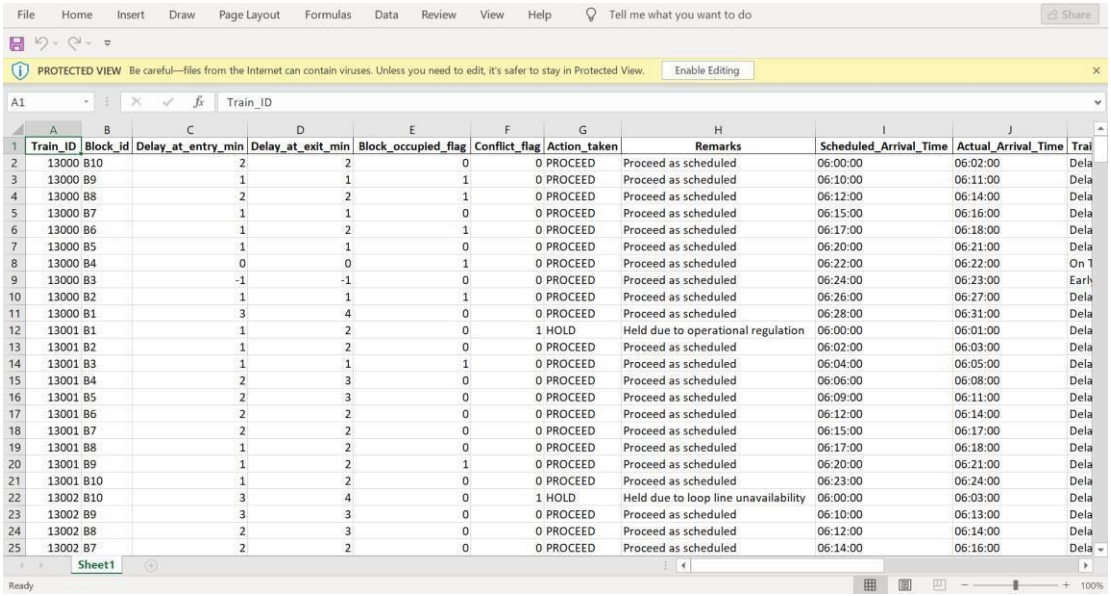
The dataset was created by integrating information from multiple railway-related sources to form a comprehensive representation of train operations and delay behavior. The primary objective was to build a structured dataset that captures scheduling details, operational characteristics, environmental factors, and performance measures required for delay prediction.

The TRAIN table serves as the core structure of the dataset. It includes essential identifiers such as *train number, train name, zone,* and station codes, which allow unique mapping of each service and its route. Temporal attributes, including *departure time, scheduled arrival, and actual arrival,* form the basis for calculating delay or early arrival values. Duration in hours and minutes was extracted from the raw time fields to support feature engineering involving travel time and section occupancy.

To represent the operational nature of each train, additional attributes were included, such as *type*, *distance category*, and coach-specific boolean fields (sleeper, chair car, AC classes). These fields help classify trains into priority groups and analyze how their physical configuration influences delay patterns. Route-related information, such as *geometry*, further supports modelling congestion and spatial movement.

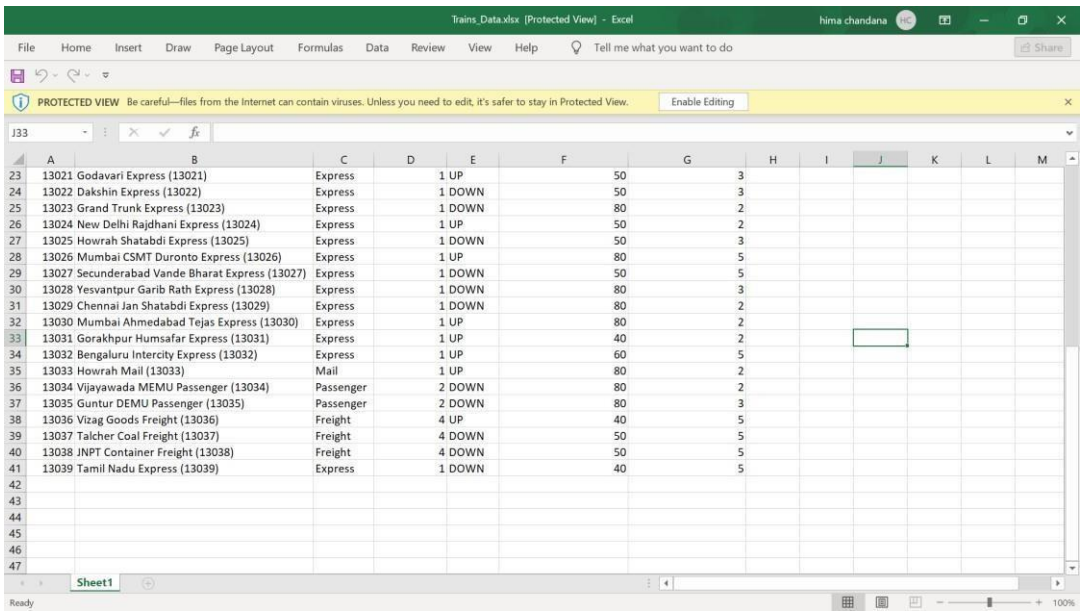
Environmental and contextual features were merged into the dataset to account for external delay factors. Weather conditions, temperature, and state were added to capture climatic influences, while *priority level* and *max permissible speed* were introduced to quantify scheduling importance and speed constraints. Finally, the *delay_or_early_in_m* field was computed to provide the target variable for machine learning models.

This dataset creation process ensured that the final structured table contains a balanced combination of temporal, operational, geographic, and environmental data, enabling effective training and evaluation of machine-learning algorithms for train delay prediction.



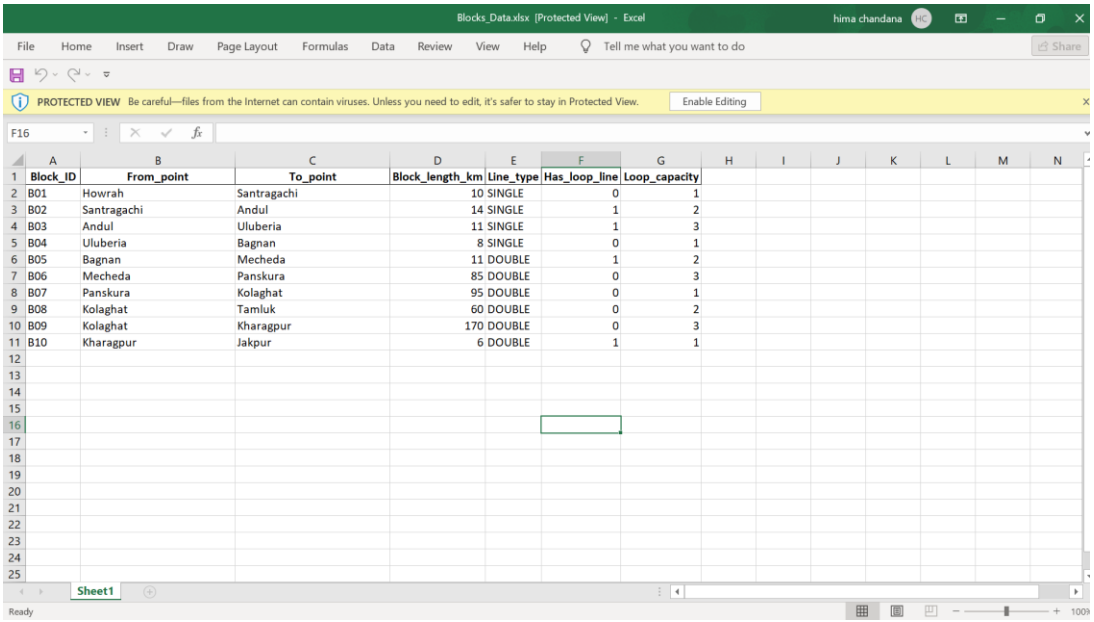
Train_ID	Block_id	Delay_at_entry_min	Delay_at_exit_min	Block_occupied_flag	Conflict_flag	Action_taken	Remarks	Scheduled_Arrival_Time	Actual_Arrival_Time	Trai
13000	B10	2	2	0	0	PROCEED	Proceed as scheduled	06:00:00	06:02:00	Dela
13000	B9	1	1	1	0	PROCEED	Proceed as scheduled	06:10:00	06:11:00	Dela
13000	B8	2	2	1	0	PROCEED	Proceed as scheduled	06:12:00	06:14:00	Dela
13000	B7	1	1	0	0	PROCEED	Proceed as scheduled	06:15:00	06:16:00	Dela
13000	B6	1	2	1	0	PROCEED	Proceed as scheduled	06:17:00	06:18:00	Dela
13000	B5	1	1	0	0	PROCEED	Proceed as scheduled	06:20:00	06:21:00	Dela
13000	B4	0	0	1	0	PROCEED	Proceed as scheduled	06:22:00	06:22:00	On T
13000	B3	-1	-1	0	0	PROCEED	Proceed as scheduled	06:24:00	06:23:00	Earl
13000	B2	1	1	1	0	PROCEED	Proceed as scheduled	06:26:00	06:27:00	Dela
13000	B1	3	4	0	0	PROCEED	Proceed as scheduled	06:28:00	06:31:00	Dela
13001	B1	1	2	0	1	HOLD	Held due to operational regulation	06:00:00	06:01:00	Dela
13001	B2	1	2	0	0	PROCEED	Proceed as scheduled	06:02:00	06:03:00	Dela
13001	B3	1	1	1	0	PROCEED	Proceed as scheduled	06:04:00	06:05:00	Dela
13001	B4	2	3	0	0	PROCEED	Proceed as scheduled	06:06:00	06:08:00	Dela
13001	B5	2	3	0	0	PROCEED	Proceed as scheduled	06:09:00	06:11:00	Dela
13001	B6	2	2	0	0	PROCEED	Proceed as scheduled	06:12:00	06:14:00	Dela
13001	B7	2	2	0	0	PROCEED	Proceed as scheduled	06:15:00	06:17:00	Dela
13001	B8	1	2	0	0	PROCEED	Proceed as scheduled	06:17:00	06:18:00	Dela
13001	B9	1	2	1	0	PROCEED	Proceed as scheduled	06:20:00	06:21:00	Dela
13001	B10	1	2	0	0	PROCEED	Proceed as scheduled	06:23:00	06:24:00	Dela
13002	B10	3	4	0	1	HOLD	Held due to loop line unavailability	06:00:00	06:03:00	Dela
13002	B9	3	3	0	0	PROCEED	Proceed as scheduled	06:10:00	06:13:00	Dela
13002	B8	2	3	0	0	PROCEED	Proceed as scheduled	06:12:00	06:14:00	Dela
13002	B7	2	2	0	0	PROCEED	Proceed as scheduled	06:14:00	06:16:00	Dela

Figure 6.1(b): Trains Movement Dataset



13021	Godavari Express (13021)	Express	1	UP	50	3
13022	Dakshin Express (13022)	Express	1	DOWN	50	3
13023	Grand Trunk Express (13023)	Express	1	DOWN	80	2
13024	New Delhi Rajdhani Express (13024)	Express	1	UP	50	2
13025	Howrah Shatabdi Express (13025)	Express	1	DOWN	50	3
13026	Mumbai CSMT Duronto Express (13026)	Express	1	UP	80	5
13027	Secunderabad Vande Bharat Express (13027)	Express	1	DOWN	50	5
13028	Yesvantpur Garib Rath Express (13028)	Express	1	DOWN	80	3
13029	Chennai Jan Shatabdi Express (13029)	Express	1	DOWN	80	2
13030	Mumbai Ahmedabad Tejas Express (13030)	Express	1	UP	80	2
13031	Gorakhpur Humsafar Express (13031)	Express	1	UP	40	2
13032	Bengaluru Intercity Express (13032)	Express	1	UP	60	5
13033	Howrah Mail (13033)	Mail	1	UP	80	2
13034	Vijayawada MEMU Passenger (13034)	Passenger	2	DOWN	80	2
13035	Guntur DEMU Passenger (13035)	Passenger	2	DOWN	80	3
13036	Vizag Goods Freight (13036)	Freight	4	UP	40	5
13037	Talcher Coal Freight (13037)	Freight	4	DOWN	50	5
13038	JNPT Container Freight (13038)	Freight	4	DOWN	50	5
13039	Tamil Nadu Express (13039)	Express	1	DOWN	40	5

Figure 6.1(c): Trains Dataset



Block_ID	From_point	To_point	Block_length_km	Line_type	Has_loop_line	Loop_capacity
B01	Howrah	Santragachi	10	SINGLE	0	1
B02	Santragachi	Andul	14	SINGLE	1	2
B03	Andul	Uluberia	11	SINGLE	1	3
B04	Uluberia	Bagnan	8	SINGLE	0	1
B05	Bagnan	Mecheda	11	DOUBLE	1	2
B06	Mecheda	Panskura	85	DOUBLE	0	3
B07	Panskura	Kolaghat	95	DOUBLE	0	1
B08	Kolaghat	Tamluk	60	DOUBLE	0	2
B09	Kolaghat	Kharagpur	170	DOUBLE	0	3
B10	Kharagpur	Jakpur	6	DOUBLE	1	1

Figure 6.1(d): Blocks Dataset

6.2. FEATURE ENGINEERING

Several domain-specific features were created to improve the model's understanding of real railway operations.

- **Priority Level**

  Trains were assigned a numeric priority based on their category (e.g., Rajdhani > Superfast > Passenger). This reflects how high-priority trains are cleared first during congestion, directly influencing delay behavior.

- **Max Speed (kmph)**

  Each train was mapped to its permissible maximum speed. Faster trains have better recovery potential and lower chances of delay, making this an important performance indicator.

- **Line Capacity**

  Line capacity was estimated based on track type (single/double), traffic density, and section characteristics. Higher-capacity routes face fewer bottlenecks, so this helps model congestion-related delays.

- **Gradient**

  Gradient values represent the slope of the route. Steeper gradients reduce speed and increase delay probability. Adding this feature helps capture terrain-based speed variations.

- **Average Block Time**

  Average block time was derived from section length and allowable speed. Longer block occupancy increases delay propagation, making it a key factor for modelling congestion.

- **Loop Lines**

  A feature indicating the availability of loop lines was added. Sections with more loop lines allow overtakes and smoother traffic flow, reducing operational delays.

6.3 Model Development

The model development stage involved training two machine-learning algorithms: Logistic Regression and Random Forest, to predict train delays and compare their performance. Both models were selected because they represent two different types of learning approaches: one linear and interpretable, the other non-linear and ensemble-based.

- **Logistic Regression**

  Logistic Regression was used as a baseline classification model to predict whether a train would be delayed or on time. It works by learning a linear relationship between the input features and the probability of delay. Because it is simple and interpretable, it helps establish a reference level of accuracy and provides insights into which features have the strongest impact on delay outcomes.

- **Random Forest**

  Random Forest, an ensemble of decision trees, was used to capture complex, non-linear patterns in the data. It is particularly effective for structured datasets where delays can depend on multiple interacting factors such as weather, priority, and congestion. The model combines predictions from many trees, improving robustness and reducing overfitting. This often leads to higher accuracy compared to simpler algorithms.

- **Model Comparison and Accuracy Evaluation**

  Both models were trained using the same training dataset and evaluated using the same test dataset. Performance metrics such as Accuracy, Precision, Recall, F1-Score, MAE, and RMSE were used to compare how well each model predicts delay behaviour. Logistic Regression provides a good baseline, while Random Forest typically offers stronger predictive performance due to its ability to handle non-linear relationships and feature interactions. Comparing both models helps determine which algorithm is more suitable for reliable delay prediction in real-world railway settings.

6.4 Training & Testing

- We will split the data into 80:20 for training and testing
- We will perform hyperparameter tuning to increase accuracy.
- Accuracy metrics such as RMSE and R^2 will be used to validate results.

Chapter 7 – DESIGN ARCHITECTURE

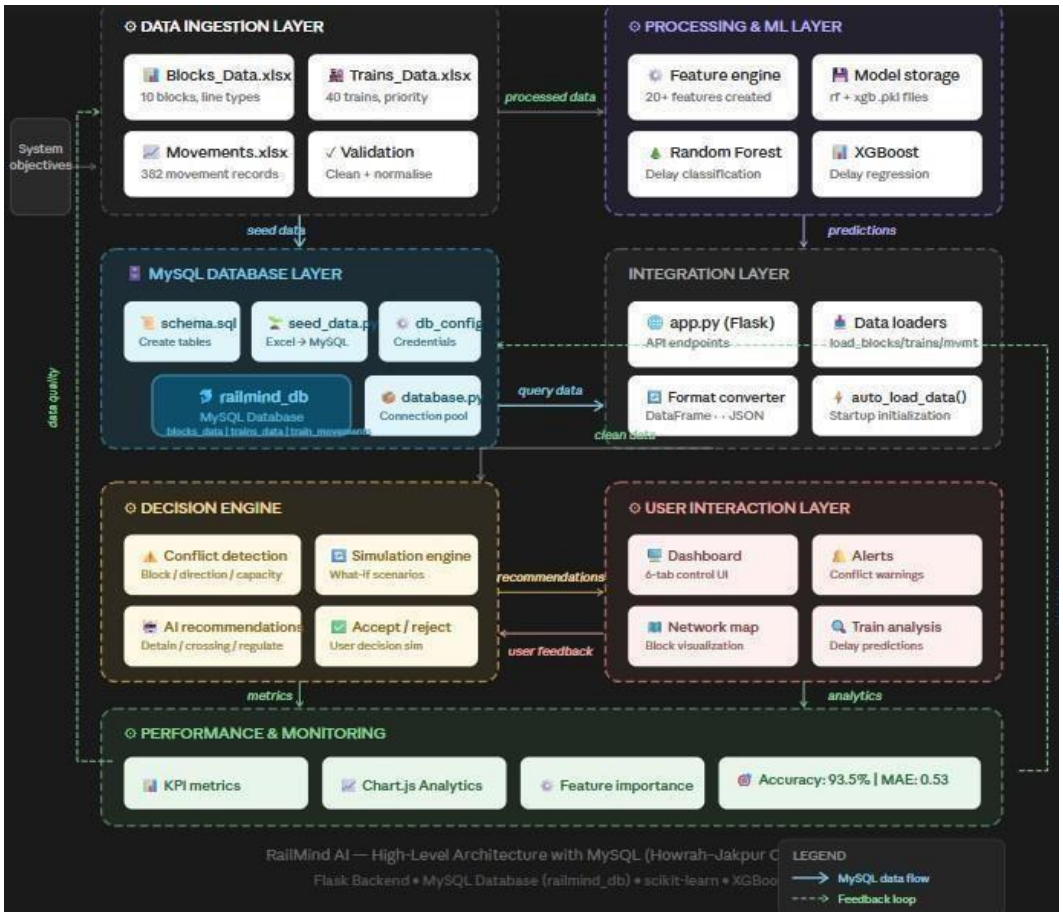


Figure 7(a): Design Architecture

The system architecture of RailMind AI consists of six major layers, each performing a critical function in processing railway operational data and generating actionable insights for section controllers along the Howrah–Jakpur corridor.

1. **Data Ingestion Layer**

   This layer handles the initial intake of raw railway data from three Excel files. Blocks_Data.xlsx contains information about 10 railway blocks (B1–B10), including station names, block lengths, line types (SINGLE/DOUBLE), and loop availability. Trains_Data.xlsx stores details of 40 trains with their names, types (Express/Passenger/Freight/Mail), priority levels, directions, and speed specifications. Train_Movements_Balanced.xlsx holds 382 historical movement records capturing delays, conflicts, actions taken, and scheduled versus actual timings. The validation component cleans and normalises all incoming data by handling missing values, standardising block ID formats (e.g., B01 → B1), and ensuring data consistency before passing it to the database layer.

2. **Processing and ML Layer**

This layer transforms raw data into intelligent predictions using machine learning algorithms. The Feature Engine creates 20+ engineered features from the cleaned data, including delay ratios, block occupancy rates, time-based features (hour of day, day of week), and categorical encodings for train types and directions. These features feed into two trained models — a Random Forest Classifier that predicts whether a train will be delayed or on time with 93.5% accuracy, and an XGBoost Regressor that predicts the exact delay duration in minutes with a Mean Absolute Error of just 0.53 minutes. Both trained models are serialised and stored as .pkl files (rf_model.pkl and xgb_model.pkl) for quick loading during application startup.

3. **MySQL Database Layer**

This layer provides persistent, structured storage, replacing the original Excel-based data reading approach. The schema.sql file defines the database structure by creating the railmind_db_2 database with three normalised tables — blocks_data, trains_data, and train_movements — complete with primary keys, foreign key relationships, and indexes for optimised query performance. The seed_data.py script performs a one-time migration of all Excel data into MySQL, handling foreign key constraints by temporarily disabling checks during bulk inserts. The db_config.py file stores database connection credentials (host, port, user, password), while database.py implements a connection pool with helper functions like load_blocks_data(), load_trains_data(), and load_train_movements() that return pandas DataFrames for seamless integration with the existing application logic.

4. **Integration Layer**

This layer acts as the bridge between the database and application components, ensuring smooth data flow across the system. The app.py Flask application serves as the central hub, exposing RESTful API endpoints for the frontend dashboard while orchestrating all backend operations. Data loader functions query the MySQL database and convert results into pandas DataFrames that the ML models and detection algorithms can process. The format converter handles bidirectional transformation between DataFrame objects used internally and JSON responses required by the web interface. The auto_load_data() function executes during application startup, automatically initialising database connections, loading all datasets into memory, training ML models, detecting conflicts, and generating recommendations — ensuring the system is fully operational the moment the server starts.

5. **Decision Engine**

This layer forms the intelligent core of RailMind AI, responsible for detecting conflicts and generating actionable recommendations. The Conflict Detection module analyzes train movements across blocks to identify three types of conflicts — block conflicts (two trains in the same block), direction conflicts (opposing trains on single-line sections), and capacity conflicts (loop line overflow). The AI Recommendations engine generates resolution strategies based on conflict type, block characteristics, and train priorities, suggesting actions like DETAIN (hold lower-priority train), CROSSING (use loop line for train crossing), or REGULATE (adjust speed/timing). The Simulation Engine enables what-if analysis by modeling hypothetical scenarios such as extra halts, speed restrictions, or route diversions. The Accept/Reject module allows controllers to evaluate and act upon recommendations, with their decisions feeding back into the system for continuous improvement.

6. **User Interaction Layer**

This layer provides the visual interface through which section controllers interact with the RailMind AI system. The Dashboard presents a 6-tab control interface built with HTML, CSS, and JavaScript, offering comprehensive views of system status, network conditions, and train operations. The Network Map tab visualizes all 10 blocks with color-coded indicators showing line types, current occupancy, and conflict hotspots. The Train Analysis tab displays individual train details with delay predictions, conflict involvement, and AI-generated recommendations specific to each train. The Alerts component provides conflict warnings with severity levels, affected trains, and suggested resolutions, ensuring controllers can respond promptly to emerging situations. All interface elements communicate with the Flask backend via AJAX calls, receiving JSON responses that Chart.js renders into interactive visualizations.

7. **Performance and Monitoring**

This layer continuously tracks system health and model effectiveness to ensure reliable operations. KPI Metrics monitors key performance indicators, including overall punctuality rate, average delay reduction, conflict resolution success rate, and recommendation acceptance percentage. Chart.js Analytics generate visual reports showing delay distribution by hour, conflicts by block, train type performance comparisons, and historical trend analysis. The Feature Importance component displays which input features contribute most to model predictions, helping identify critical factors affecting train delays. Model Performance tracking validates that the Random Forest maintains its 93.5% classification accuracy and XGBoost stays within the 0.53-minute MAE threshold, triggering alerts if performance degrades. Feedback loops from this layer flow back to both the Data Ingestion Layer (for data quality improvements) and the ML Layer (for model retraining triggers), creating a self-improving system architecture.

Chapter 8 – RESULTS

System Overview Dashboard

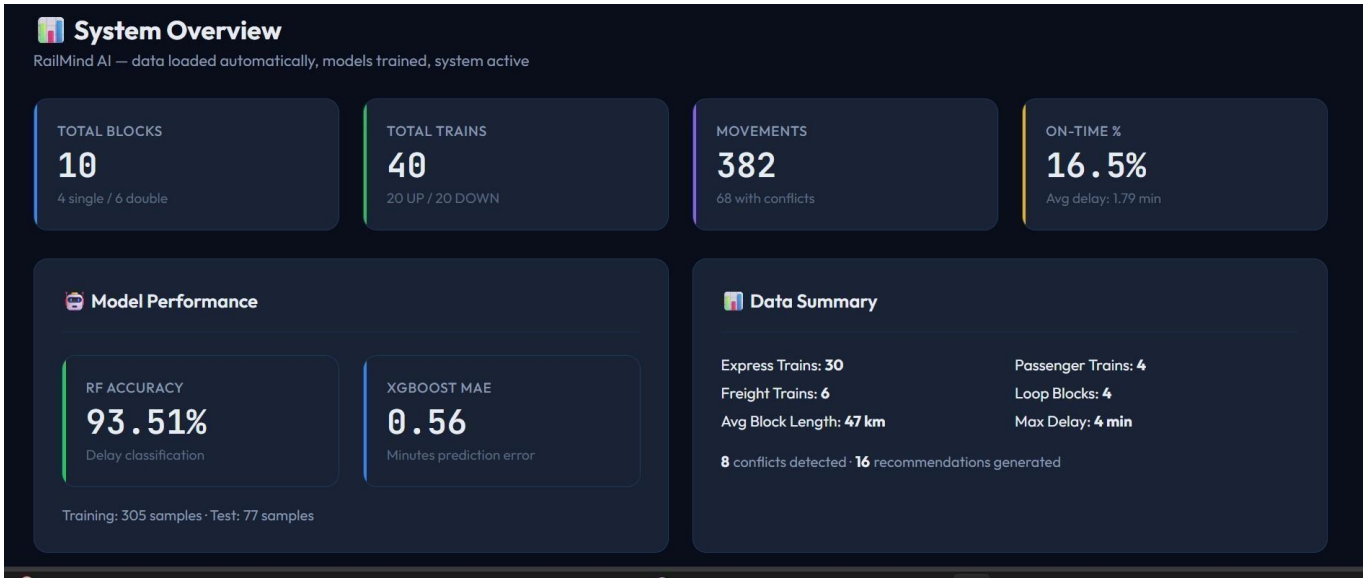


Figure 8(a)

Railway Network Overview

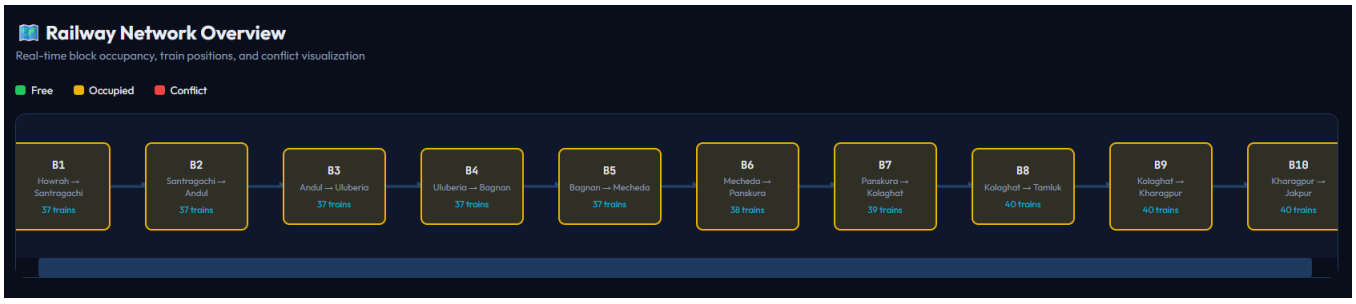


Figure 8(b)

Train Analysis Dashboard

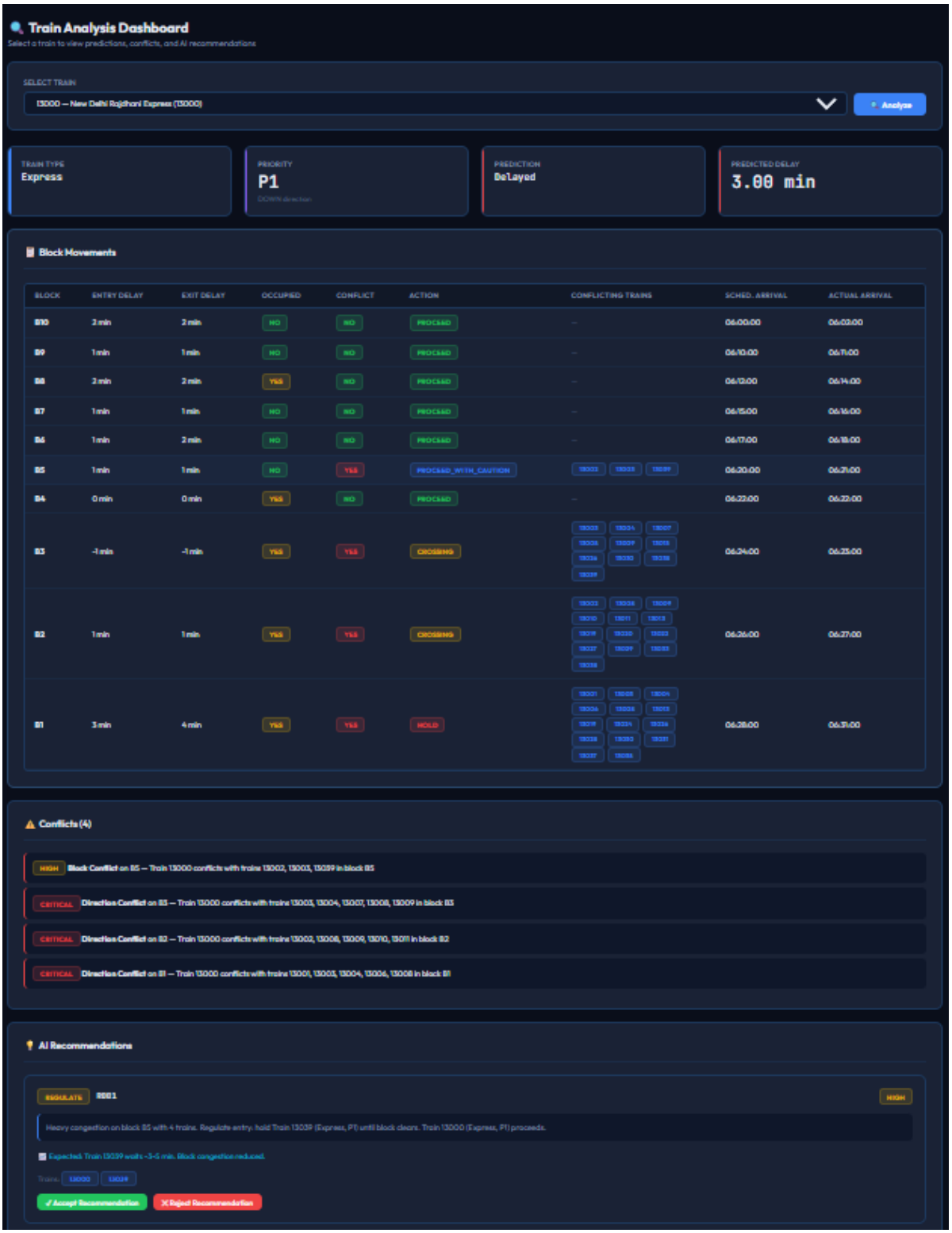
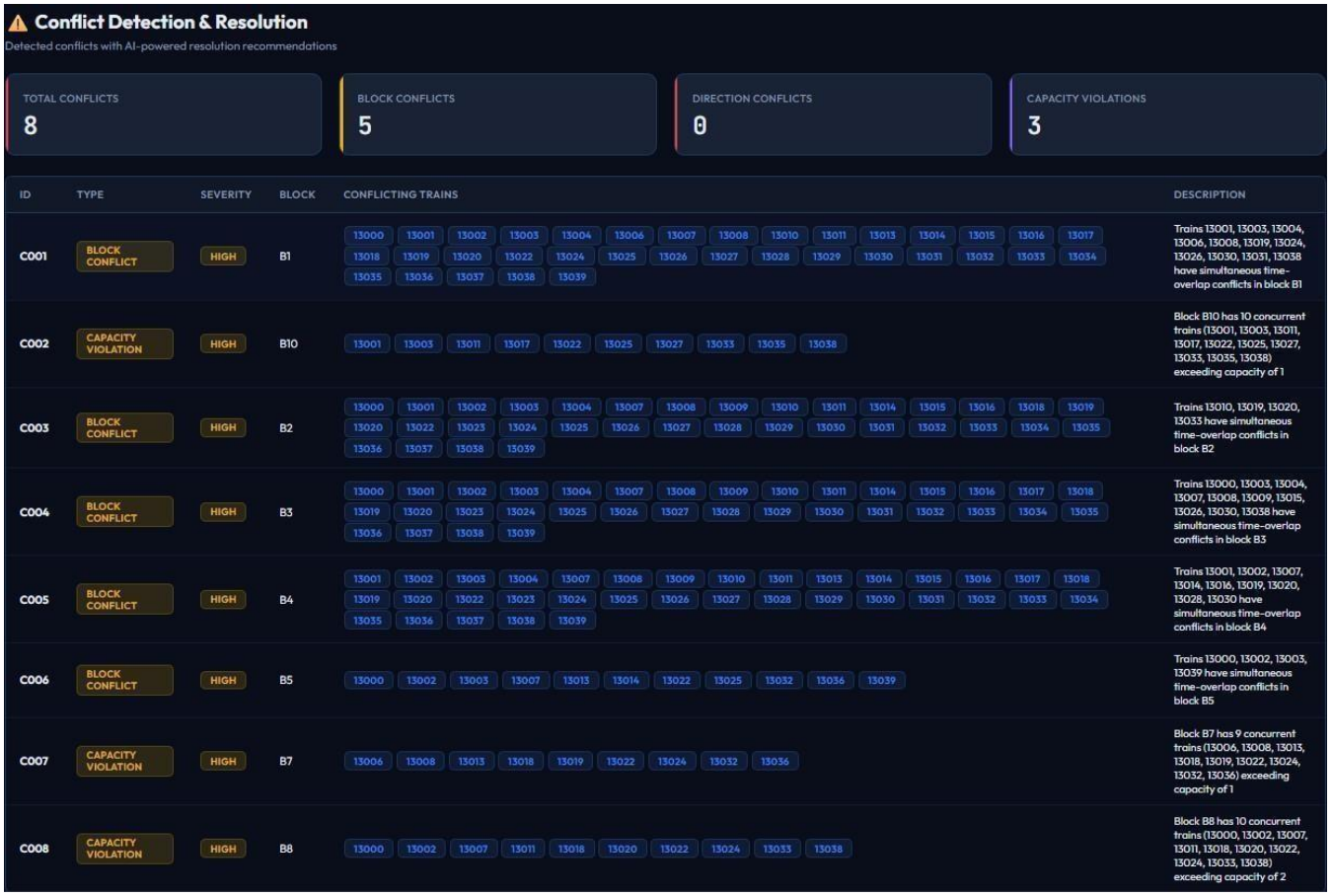


Figure 8(c)

Conflicts Dashboard



Conflict Detection & Resolution

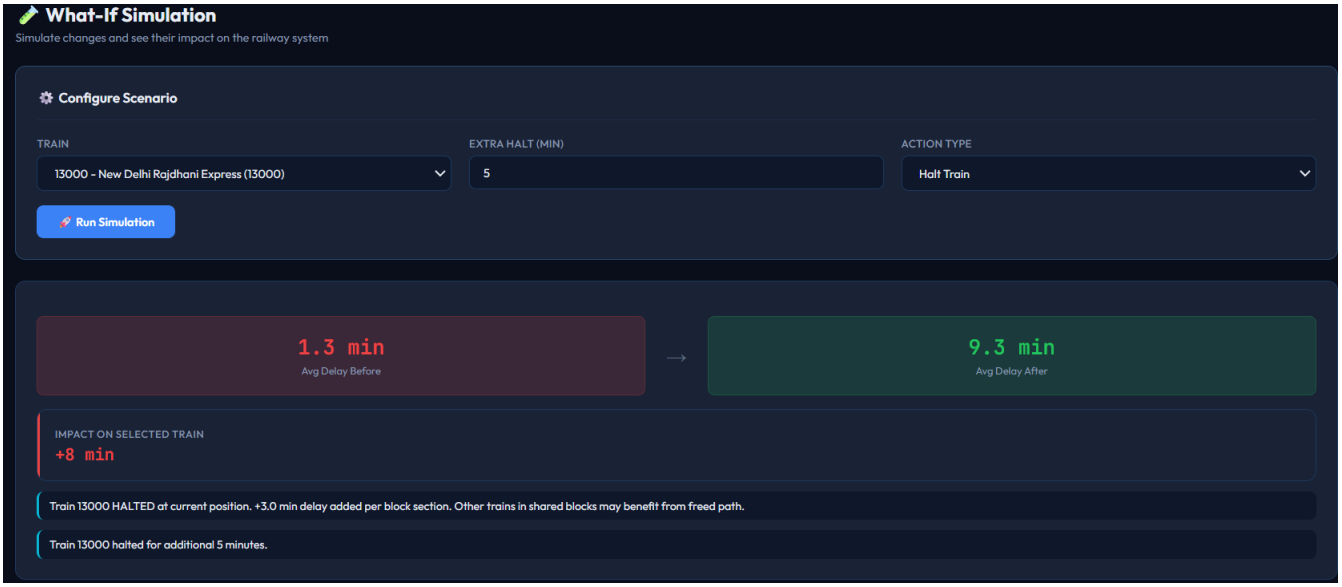
Detected conflicts with AI-powered resolution recommendations

TOTAL CONFLICTS	BLOCK CONFLICTS	DIRECTION CONFLICTS	CAPACITY VIOLATIONS
8	5	0	3

ID	TYPE	SEVERITY	BLOCK	CONFLICTING TRAINS	DESCRIPTION
C001	BLOCK CONFLICT	HIGH	B1	13000 13001 13002 13003 13004 13006 13007 13008 13010 13011 13013 13014 13015 13016 13017 13018 13019 13020 13022 13024 13025 13026 13027 13028 13029 13030 13031 13032 13033 13034 13035 13036 13037 13038 13039	Trains 13001, 13003, 13004, 13006, 13008, 13019, 13024, 13026, 13030, 13031, 13038 have simultaneous time-overlap conflicts in block B1
C002	CAPACITY VIOLATION	HIGH	B10	13001 13003 13011 13017 13022 13025 13027 13033 13035 13038	Block B10 has 10 concurrent trains (13001, 13003, 13011, 13017, 13022, 13025, 13027, 13033, 13035, 13038) exceeding capacity of 1
C003	BLOCK CONFLICT	HIGH	B2	13000 13001 13002 13003 13004 13007 13008 13009 13010 13011 13014 13015 13016 13018 13019 13020 13022 13023 13024 13025 13026 13027 13028 13029 13030 13031 13032 13033 13034 13035 13036 13037 13038 13039	Trains 13010, 13019, 13020, 13033 have simultaneous time-overlap conflicts in block B2
C004	BLOCK CONFLICT	HIGH	B3	13000 13001 13002 13003 13004 13007 13008 13009 13010 13011 13014 13015 13016 13017 13018 13019 13020 13023 13024 13025 13026 13027 13028 13029 13030 13031 13032 13033 13034 13035 13036 13037 13038 13039	Trains 13000, 13003, 13004, 13007, 13008, 13009, 13015, 13026, 13030, 13038 have simultaneous time-overlap conflicts in block B3
C005	BLOCK CONFLICT	HIGH	B4	13001 13002 13003 13004 13007 13008 13009 13010 13011 13013 13014 13015 13016 13017 13018 13019 13020 13022 13023 13024 13025 13026 13027 13028 13029 13030 13031 13032 13033 13034 13035 13036 13037 13038 13039	Trains 13001, 13002, 13007, 13014, 13016, 13019, 13020, 13028, 13030 have simultaneous time-overlap conflicts in block B4
C006	BLOCK CONFLICT	HIGH	B5	13000 13002 13003 13007 13013 13014 13022 13025 13032 13036 13039	Trains 13000, 13002, 13003, 13039 have simultaneous time-overlap conflicts in block B5
C007	CAPACITY VIOLATION	HIGH	B7	13006 13008 13013 13018 13019 13022 13024 13032 13036	Block B7 has 9 concurrent trains (13006, 13008, 13013, 13018, 13019, 13022, 13024, 13032, 13036) exceeding capacity of 1
C008	CAPACITY VIOLATION	HIGH	B8	13000 13002 13007 13011 13018 13020 13022 13024 13033 13038	Block B8 has 10 concurrent trains (13000, 13002, 13007, 13011, 13018, 13020, 13022, 13024, 13033, 13038) exceeding capacity of 2

Figure 8(d)

Simulation Dashboard



What-If Simulation

Simulate changes and see their impact on the railway system

Configure Scenario

TRAIN: 13000 - New Delhi Rajdhani Express (13000)

EXTRA HALT (MIN): 5

ACTION TYPE: Halt Train

Run Simulation

1.3 min
Avg Delay Before

→

9.3 min
Avg Delay After

IMPACT ON SELECTED TRAIN
+8 min

Train 13000 HALTED at current position. +3.0 min delay added per block section. Other trains in shared blocks may benefit from freed path.

Train 13000 halted for additional 5 minutes.

Figure 8(e)

Performance Dashboard

Figure 8(f)

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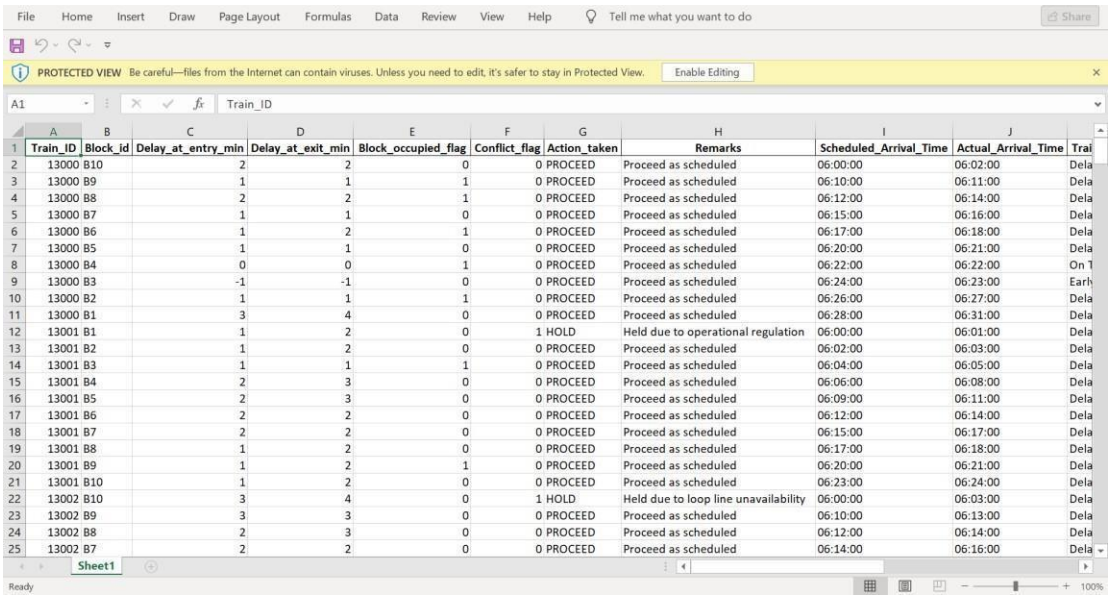
[12] Y. Liu, P. Sun, J. Zhang and Z. Jiang, “Railway Delay Propagation Prediction Based on Spatio-Temporal Graph Neural Networks,” *Applied Soft Computing*, vol. 122, p. 108889, 2022.

APPENDIX

Dataset Snapshot:



Train_ID	Block_id	Delay_at_entry_min	Delay_at_exit_min	Block_occupied_flag	Conflict_flag	Action_taken	Remarks	Scheduled_Arrival_Time	Actual_Arrival_Time	Trai
13000	B10	2	2	0	0	PROCEED	Proceed as scheduled	06:00:00	06:02:00	Dela
13000	B9	1	1	1	0	PROCEED	Proceed as scheduled	06:10:00	06:11:00	Dela
13000	B8	2	2	1	0	PROCEED	Proceed as scheduled	06:12:00	06:14:00	Dela
13000	B7	1	1	0	0	PROCEED	Proceed as scheduled	06:15:00	06:16:00	Dela
13000	B6	1	2	1	0	PROCEED	Proceed as scheduled	06:17:00	06:18:00	Dela
13000	B5	1	1	0	0	PROCEED	Proceed as scheduled	06:20:00	06:21:00	Dela
13000	B4	0	0	1	0	PROCEED	Proceed as scheduled	06:22:00	06:22:00	On T
13000	B3	-1	-1	0	0	PROCEED	Proceed as scheduled	06:24:00	06:23:00	Earl
13000	B2	1	1	1	0	PROCEED	Proceed as scheduled	06:26:00	06:27:00	Dela
13000	B1	3	4	0	0	PROCEED	Proceed as scheduled	06:28:00	06:31:00	Dela
13001	B1	1	2	0	1	HOLD	Held due to operational regulation	06:00:00	06:01:00	Dela
13001	B2	1	2	0	0	PROCEED	Proceed as scheduled	06:02:00	06:03:00	Dela
13001	B3	1	1	1	0	PROCEED	Proceed as scheduled	06:04:00	06:05:00	Dela
13001	B4	2	3	0	0	PROCEED	Proceed as scheduled	06:06:00	06:08:00	Dela
13001	B5	2	3	0	0	PROCEED	Proceed as scheduled	06:09:00	06:11:00	Dela
13001	B6	2	2	0	0	PROCEED	Proceed as scheduled	06:12:00	06:14:00	Dela
13001	B7	2	2	0	0	PROCEED	Proceed as scheduled	06:15:00	06:17:00	Dela
13001	B8	1	2	0	0	PROCEED	Proceed as scheduled	06:17:00	06:18:00	Dela
13001	B9	1	2	1	0	PROCEED	Proceed as scheduled	06:20:00	06:21:00	Dela
13001	B10	1	2	0	0	PROCEED	Proceed as scheduled	06:23:00	06:24:00	Dela
13002	B10	3	4	0	1	HOLD	Held due to loop line unavailability	06:00:00	06:03:00	Dela
13002	B9	3	3	0	0	PROCEED	Proceed as scheduled	06:10:00	06:13:00	Dela
13002	B8	2	3	0	0	PROCEED	Proceed as scheduled	06:12:00	06:14:00	Dela
13002	B7	2	2	0	0	PROCEED	Proceed as scheduled	06:14:00	06:16:00	Dela

Figure A.1: Dataset Snapshot

Recommendation System's ER Diagram:

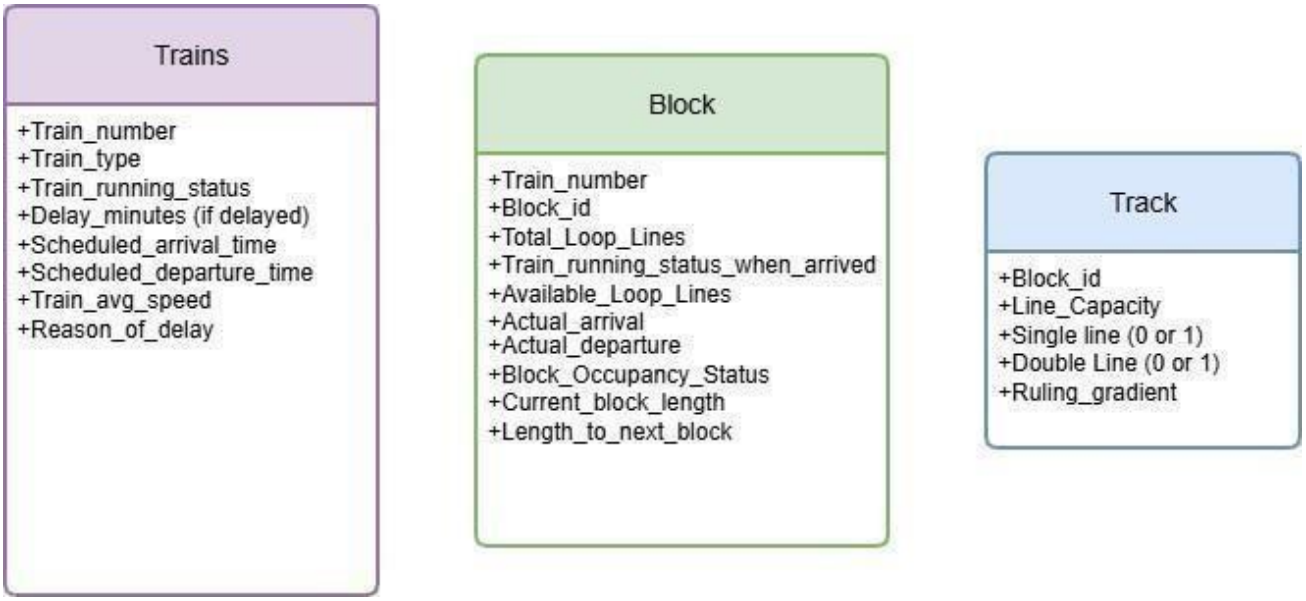


Figure A.2: Recommendation System

RESEARCH PAPER

Comparative study based on machine learning train delay prediction for Indian Railways

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***Abstract*— Indian Railways' net, one of the greatest passenger and freight networks in the world, is now highly dependent on manual decision-making and the gut instinct of Train Traffic Controllers. With the high frequency and density of trains, this manual method has serious limitations, causing a chain reaction of delays, high network load, and low efficiency. The system takes raw movement information, which includes scheduling metadata, sectional properties, line capacities, priority types, and weather conditions, to predict future delays of the train and find the block occupancy time. It consists of five layers of the technical methodology, each consisting of foundational data (Input Layer), data cleaning and route graph construction (Pre-processing Layer), the use of the Random Forest algorithm (Prediction Layer), occupancy analysis (Simulation Layer) and actionable reporting (Visualisation Layer). Experimental findings have shown that the ensemble Random Forest model is better than linear models at capturing the complex, non-linear relationship of the network. The completed plan is to combine all these predictive models into an interactive analytics dashboard, which visualises heatmaps and occupancy schedules, and sends out warnings in case there are high-congestion levels. This system will help Traffic Controllers to provide data-based, algorithmic advice aimed at enhancing the punctuality, safety, and overall responsiveness of the railway traffic management.**

Keywords: Congestion Forecasting, Dashboard Visualisation, Indian Railways, Machine Learning, Operational Efficiency, Real Time Alerts Section, Traffic Management, Train Delay Forecasting.

I. INTRODUCTION

The efficiency of the rail system in a country is usually seen as a measure of its economic well-being and the development of its infrastructure. Railway transport is a cyber-physical system that facilitates the large-scale movement of passengers and logistics of the supply chain in the industry. Such networks, however, are prone to disruptions. Even with the high level of sophistication of systems, it remains impossible to have the "perfect schedule" due to the extremely unpredictable circumstances under which the trains operate. Delays are no longer an isolated phenomenon, but systematic symptoms brought by the combination of mechanical reliability, environmental fluctuities and human working constraints. Once the first delay occurs, which is defined as the deviation of the timetable, it rarely remains individual. Instead, it triggers a sequence of delays termed as knock-on delays in which the usual usage of tracks, platforms and signalling segments causes the latency to propagate across the entire geographic layout of the network.

There are economic and social consequences of these postponements. To the travellers, there are varying travel periods leading to inferior quality and low efficiency. In the case of manufacturing industries, the insecurity in the rail system encourages a shift to road transportation, which increases the effect of freight and aggravates traffic on major highways. Traditionally, these problems have been addressed by the railway authorities through the use of buffer times, which a periods of slack time built into timetables to cater to delays. This approach is efficient, but is, in the most fundamental sense, inefficient as it does not exploit the physical potential of the track. The demerits of these fixed reactive adjustments are felt more during peak seasons or extreme weather conditions, where the buffer empties very fast, disrupting the whole schedule.

The essence of the problem that will be addressed in the study is the direct transformation of the traditional reactive model into a proactive data-driven predictive system. The recent developments in intelligence, added to the general digitisation of railway records (via SCADA and IoT-based telemetry), have created a unique opportunity to apply supervised machine learning to work with practical transit physics. Using the railway logs as feature spaces, we can then move beyond simple descriptive statistics in order to develop more complex mapping functions that would allow relating specific variables such as line capacity, track density and weather visibility with probable delay outcomes.

The methodological approach introduced in this research the possibility to forecast operational statuses (Early On-time or delayed) and the possibility to estimate the exact time deviation in the number of minutes. We consider three distinct modelling models, namely Logistic Regression, which establishes a baseline with the help of sigmoid-based probability; Random Forest, which utilises a cluster of independent decision trees to handle high-variance data; and XGBoost, which uses gradient descent in a functional space to improve regression accuracy. One of the main aspects of this introduction is the focus on limitations. Whereas ordinary intuition tends to attribute sluggishness to factors like weather, this paper examines the hypothesis that it is the

physical constraints of the network, such as the blockage of loop lines and the constraints of line capacity, that are the actual invisible cause of unreliability. By quantifying these impacts, the study aims to provide the planners in the railway sector with a prediction tool in addition to a diagnostic model to concentrate on infrastructure investments. The following passages outline the strict data engineering, mathematical modelling and validation of these systemic results.

II. LITERATURE SURVEY

Traditionally, Indian Railways has been a massive transportation network in the world, whose operations are based on deterministic principles. The scheduling was a strict mathematical limit handled by the human knowledge of the Train Traffic Controller. These human operators depended on their feelings and experience to control track sections that had a length of thousands of kilometres. Initial studies at this stage pointed out systematic analysis of statistical models but raised great concerns about generalizability and privacy. Coordination through the manual system worked out well in low train frequency cases; however, as the train frequency increased and the complexity in the operational routes increased, the small delays started to spread across the interconnected routes, generating a ripple effect. The literature has confirmed that even though traditional signalling was safe, it was not intelligent enough to forecast dynamic section occupancy.

The scholarly origin of supervised learning in this direction originated with the laid down linear standards. The adoption of Logistic Regression was because it is an easy and interpretable linear learning method. It was employed by early researchers to tabulate reference levels of accuracy and give information on features that had the strongest influence on delay. It set up a yardstick for categorising a delayed train or a timely train. Nevertheless, it has always been indicated in the literature that Logistic Regression does not deal well with factors that have interaction. In complicated railway networks where the railroad Section length, the speed of the trains, and the priority took precedence in the normal scheduling, the linear relationship learned by the Logistic Regression was the baseline for the industry to surpass.

To establish a reference point on this momentous project report the analysis will focus on the complicated transformation of the traditional Indian Railways dispatch systems to sophisticated predictive systems. Historically, the Indian Railways system has operated as a deterministic system in which Traffic Controllers have relied on manual decisions. As the density of the network rises, this reliance on instinct is often limited, where ripple delays propagate over connected block segments. Preliminarily, these ancient methods lack the necessary flexibility to handle contemporary issues. This has prompted academic consensus to shift towards data-driven systems capable of making predictions on section occupancy and computing block usage duration in order to enhance planning.

Models like Logistic Regression began as the starting point of learning in this area and developed a linear baseline of delay prediction. Logistic Regression has been appreciated due to its simplicity since it represents a linear relationship between the input variables, such as timetable metadata and segment characteristics and the probability of a train to arrive on time. Comparative studies, however, suggest that this model does provide a baseline, but this accuracy begins to plateau in situations where there the complex non-linear congestion patterns that are characteristic of the busy railway networks.

Linear models have not been able to capture these complexities; hence, researchers have sought to investigate ensemble learning-based techniques, including the Random Forest method. It has been stressed that Random Forest works well with structured data when the delays are affected by the interaction of factors like weather, priority and congestion. Random Forest helps reduce overfitting by building uncorrelated decision trees, and the results produced by the Forests are useful and can be relied upon by decision-makers. Empirical data on Operations Control Centre databases prove that predictive accuracy is higher with the use of ensemble approaches. Besides, the significance of the features derived thereof indicates that the infrastructural variables, such as line capacity and section occupancy, assume a more predictive value than the secondary variables.

The methods of modelling were more refined by using XGBoost in order to make accurate predictions of the occupancy times of sections. Researchers have noted that XGBoost is an effective tool to perform delay forecasting through regression of latencies through the boosting mechanisms. This approach uses movement information to know occupancy time to allow controllers to make contextual decisions. In modern studies, the adoption of this model approach is highlighted to support Train Controllers with algorithm-based recommendations that will contribute to a better situational awareness.

The effects of the structure of the railways, length of blocks and length of the segment remain an issue of frequent interest in research. The aspects such as the maximum speed and gradient profiles play a vital role in determining the travel time and the simulation of the train movements on different track segments. Long block sections are likely to increase the chances of delay propagation, and section occupancy development becomes an operational requirement. It has been found that loop line segments with higher track capacity have fewer congestion problems, and train overtaking can be done with significantly easier efforts. Studies on route congestion indicate that even a minor delay in a block may cause a cascading effect on the network reliability in large regions.

The modelling factors of multiclass delays in the Indian scenario are climatic. Random Forest and Artificial Neural Networks (ANN) surpass the model when the weather variables, such as mist or heavy rains, are found to be important. The contextual climatic records will capture weather effects on the train speeds and will justify the incorporation of weather-related information in real-time notification systems. This set of features and environmental factors is the contemporary model of forecasting the sectional congestion rates.

Advanced spatio-temporal behaviours have guided researchers to the use of Spatio-Temporal Graph Neural Networks (ST-GNN) in investigating network patterns. Cost-sensitive approaches view timetables as images to determine delay propagation patterns. The researchers on data-driven rescheduling emphasise the idea of the power of organised datasets to promote agility in operational environments. Operations also require immediate response time, especially

less than two seconds, to maintain situational awareness. They recommend the use of modular designs to ensure improvement of components and wide scalability of the components as the amount of data grows in the railway system.

Socio-economic effects examined in the recent surveys show that future delays predictive information has a direct positive impact on the experience of passengers and the reliability of the network. With the help of intelligent tools that complement the already existing Zenith traffic management tools, operators can control section occupancy more accurately and, therefore, reduce cascading delays. Contemporary railway analytics finds that predictions based on machine learning and evaluated as an extension of human expertise can be the most useful tool in helping traffic controllers increase throughput in the overall section.

III. METHODOLOGY AND ARCHITECTURE

This project includes a system architecture and the methodology that combine to create a data-driven pipeline that converts the raw railway operational logs into meaningful insights that can be used in traffic management. The data set was constructed by combining data from various railway sources to fully reflect the train operations and delay behaviour, the major aim of which was to develop a structured data set to capture the scheduling detail, the nature of their operation, environmental issues, and real-time performance indicators. Data should have been collected by means of access to official railway schedules, picking up the way information on railway maps and schedules, collecting historical weather information with the help of meteorological sources and operational parameters with the help of railway technical documents.

The design to be proposed has five separate functional layers, which ensure high situational awareness of operational controllers. The input layer serves as the main point of ingestion, and it uses train metadata, timetables, speed profiles, types of locomotives, weather conditions, and past congestion. The TRAIN table is the main framework of the dataset with identifiers of trains, such as train number, name, zone on the railways, and codes of the station to be uniquely mapped. The temporal feature, such as departure time, scheduled arrival and actual arrival, was generated to derive delay or early arrival. After ingestion, the pre-processing layer deals with discrepancies like missing values or duplicates to build a route graph to represent movements between block sections.

The feature engineering gives numeric priority levels depending on the type of trains, like express, passenger or freight, indicating the importance of high-priority trains being cleared first in case of congestion. Other characteristics like maximum allowed speed, presence of loop lines and gradients were incorporated to measure the scheduling significance and speed limits. The prediction layer uses machine learning models, including the random forest algorithm, to predict train delays. Although Logistic Regression determines a linear level of accuracy, as it relies on developing a linear relationship between the input variables and delay probability, Random Forest obtains the opportunities to characterise intricate, non-linear relationships between influencing variables, including weather and congestion. Gradient values are used to represent speed changes caused by the topography, since areas with steep climbs decelerate trains greatly.

To identify operational movement under various conditions in a network, the simulation layer computes the occupancy of sections and simulates the flow of trains. The potential of delay propagation is reflected in the average block time, which is a derivative of the section length and allowable speed. The availability of loop lines as an option means that there is a section in which the trains can overtake each other, and the priority trains will not be held up by the slower freight trains. Both models were tested on a shared set, and the results were compared with an 80:20 stratified split, and the results were measured by such metrics as Accuracy, F1-Score, and the Root Mean Squared Error (RMSE) to guarantee that the model is not overfitting. Lastly, these insights are combined by the visualisation layer into an interactive analytics dashboard with indicators of predicted delays and section throughput to help decision-makers. In order to maintain responsiveness that is essential in dynamic conditions, the system aims for a near real-time response of less than two seconds.

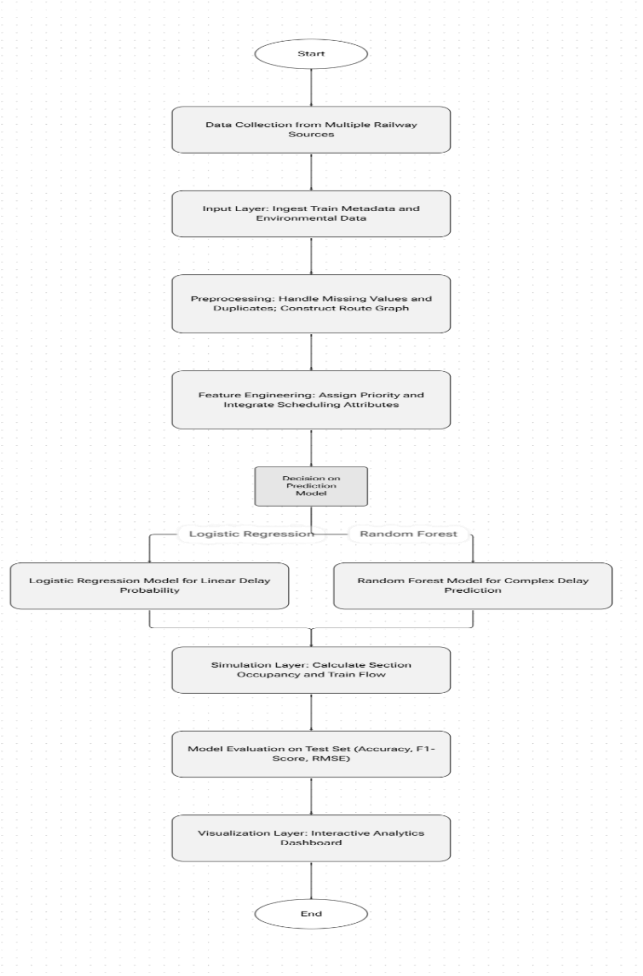

Fig 3.1 Predictive Modelling Workflow

Feature Importance Analysis and Model Validation

The Random Forest algorithm will offer scores of feature importance in relation to the extent to which each feature will be used in minimising a prediction error. These scores assist in justifying the engineering decisions and refining opportunities, and also process non-linear interactions like the U-shaped effect of temperature on delays. The infrastructural bottlenecks, such as line capacity and loop lines, as shown in our visualisations, have greater predictive weights compared to the secondary indicators.

As the comparative analysis shows, the Random Forest model achieved an approximate 0.9 (90) accuracy, and therefore met non-functional requirements, whereas the methods of Logistic Regression reach a level of lower accuracy because it is linear. A confusion matrix validation demonstrates good True Positive results with the model distinguishing between 406 delayed trains and 370 on-time arrivals, which proves that there is a high operational reliability. Moreover, the accuracy of the regression used in scatter plots indicates that forecasted latencies are close to the actual historical records, which proves the effectiveness of the system when used as an assistant dispatching tool.

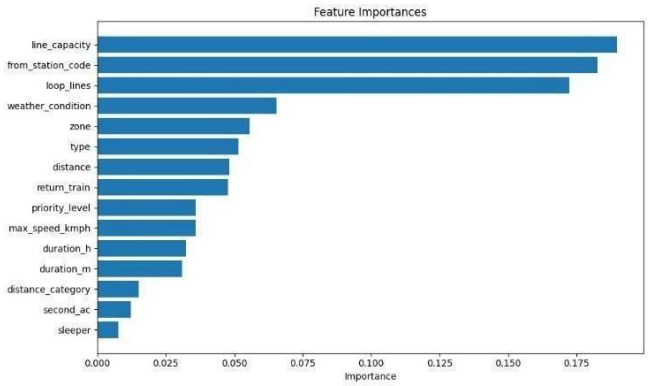

Fig 3.2 Infrastructural Priority Ranking

This horizontal bar chart displays the relative significance of the input variables to train delays prediction, as determined by the decrease in the Gini Impurity of the random forest model. The length of each bar is the contribution made to the decision-making process of the model by a particular feature. The findings indicate that there is a decisive infrastructural influence over environmental factors in the determination of operational reliability. Of all the variables, line_capacity is the most impactful, which highlights its relevance in the determination of the contribution of congestion and throughput to delay patterns. This is then followed by from station code and loop lines that both indicate structural constraints and routing complexity. Their large importance scores indicate that the physical bottlenecks are important contributors to delay propagation due to physical constraints in the form of track availability or congested junctions. Weather-related variables are, in contrast, not very important, yet are useful to contextualise disruptions. It shows that the environmental conditions can only contribute to delays, but are not key factors in the predictive model. The difference in the importance of the features justifies how the project focuses on section-level density and a geometric layout as the important levers to enhancing reliability. Thequantification of the effect of each variable allows the chart to offer practical information on the infrastructure planning and operational optimisation, leading the stakeholders to focus on track-side improvements rather than on reactive environmental adaptations.

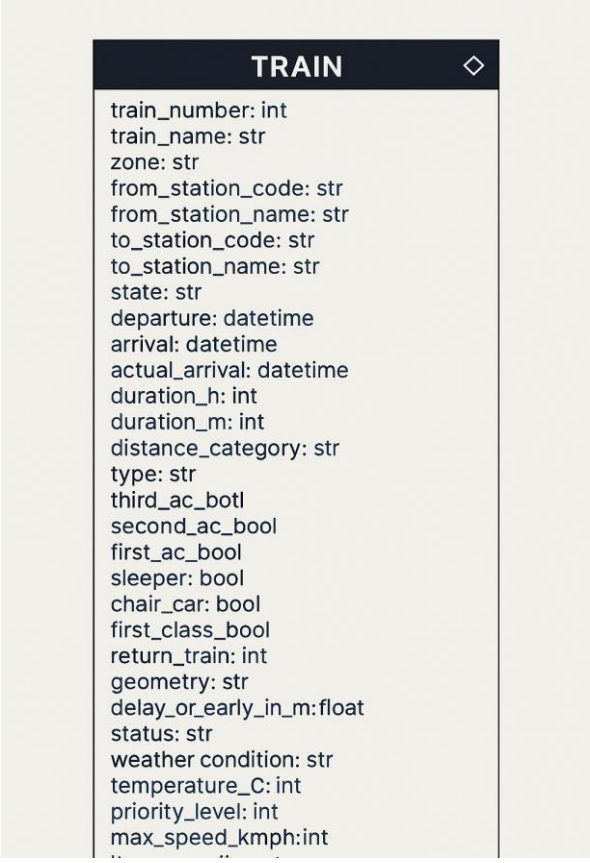

Figure 3.3 Dataset Class Diagram

The data were divided into training and test sets in 80-20 proportions, and the test set has data that has not been observed in the training. Stratified splitting was done to ensure that similar distributions of train types and delay patterns were present in both sets. To optimise the model performance, we did hyperparameter tuning by a grid search method with cross-validation. We optimised the regularisation strength of the model to balance the complexity of the model and generalisation, in the case of Logistic Regression. In the case of Random Forest, we were able to optimise the number of trees, the maximum number of trees in each tree, the maximum number of samples per leaf, and the rate of sampling of features. Cross-validation was used to split training data into several folds and then to train the model on some folds and validate on others, and then average them to come up with strong estimates. We have computed evaluation metrics to determine predictive accuracy, and we have created confusion matrices to visualise right and wrong classification. Confusion matrices demonstrate true positives, true negatives, false positives as well and false negatives, which indicate the existence of systematic biases in the model. The structured data creation, domain-specific feature engineering, and comparative model development are a full-fledged methodology of predicting train delays and helping traffic controllers make well-informed decisions based on data.

IV. RESULT

The empirical analysis of the Indian Railways train delay predictive framework is an inclusive quantitative measure of the reliability of the system in terms of multi-layered performance measures and visual diagnostics. The comparison of the performance has provided a clear baseline comparison between the applied algorithms and has shown that the ensemble Random Forest model has a consistent accuracy, precision and recall score of about 0.9(90%), thus meeting the target accuracy of the project in making reliable decisions.

The issue of classification reliability was also questioned using the heatmap of the confusion matrix that provides detailed evidence of the accuracy of the Random Forest model in discriminating between operational categories (e.g., Delays, Early, and On-time arrivals). It can be seen in the matrix that the system was able to identify 406 instances of delays and 370 instances of on-time arrival in the correct way, which proves that the system is sensitive enough to activate a real-time congestion signal to Traffic Controllers. The low level of leakage among categories means that the insights based on the data will be helpful to the decision-makers without creating an unwarranted number of false alarms.

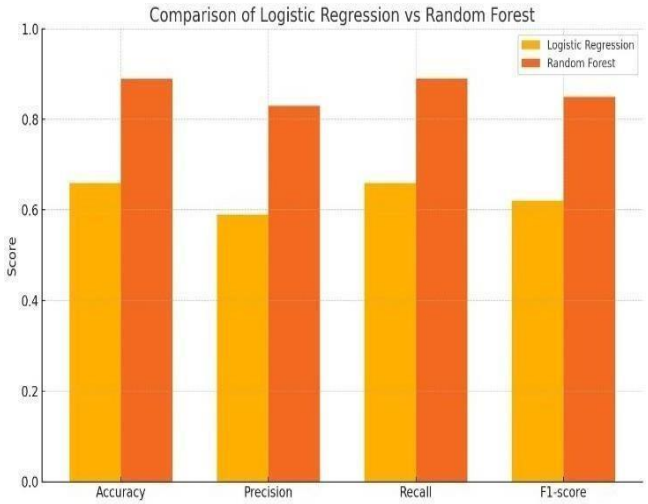

Fig 4.1 Performance Comparison Graph

This bar chart is a comparison between two classification algorithms, namely, Logistic Regression and Random Forest, with four measures of evaluation, which include: Accuracy, Precision, Recall and F1-score. General correctness is represented by Accuracy, Precision indicates the number of predicted positives that were really positive, Recall indicates the number of actual positives that were correctly predicted, and F1-score is a balance between Precision and Recall. Random Forest has better scores in all metrics in this chart, which implies that it can handle the complexity of the data better. As it is simpler and interpretable, Logistic Regression does not capture subtle patterns and can be observed by its inferior Recall and F1-score. This comparison shows a trade-off between model interpretability (Logistic Regression) and predictive power (Random Forest), and helps to choose a model based on the need to emphasise either accuracy or explainability.

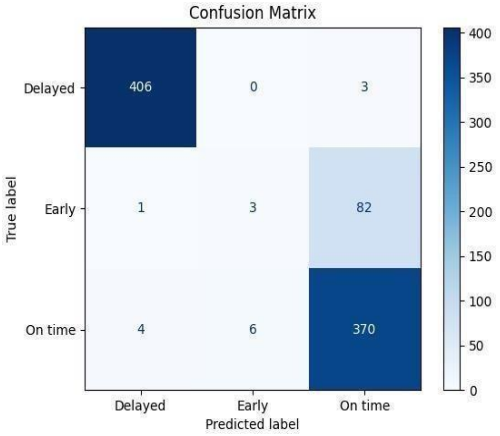

Fig 4.2 Classification Reliability Graph

The confusion matrix provides a granular view of how the classification model performs across three categories: Delayed, Early, and On time. Each row represents the true label, while each column represents the predicted label. The model performs very well for "Delayed" and "On time," with high counts of correct predictions and relatively few errors. However, the "Early" category is problematic: only 3 instances are correctly classified, while 82 are misclassified as "On time."

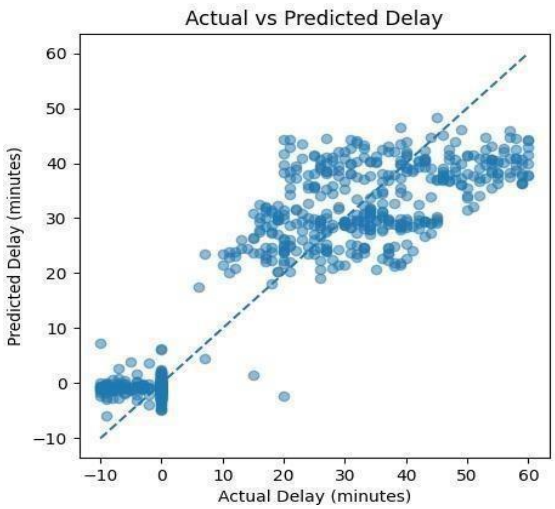

Fig 4.3 Regression Precision Graph

This scatter plot is designed to evaluate the performance of a regression model that predicts delay times. The x-axis represents the actual delay values, while the y-axis represents the predicted delay values. The diagonal dashed line serves as a reference for perfect predictions—if the model were flawless, every point would lie exactly on this line. In practice, the points cluster around the line, showing that the model captures the general trend of delays reasonably well. However, the spread of points away from the line indicates prediction errors, with some delays being underestimated or overestimated. This visualisation is particularly useful for identifying systematic bias (e.g., consistently underpredicting long delays) and spotting outliers where the model struggles.

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